Module_3: *(Template)*

Team Members:

- Kelly Kim
- Yoshiyuki Ito

Project Title:

Analyzing Breast Cancer Expression Data of Luminal Metastatic Patients

Project Goal

This project seeks to determine whether breast cancer (BRCA) tumors with higher expression of genes related to immune response (IRG) are associated with better survival outcomes in luminal metastatic patients. We will use the Cancer Genome Atlas (TCGA) and Tumor Immune Estimation Resource (TIMER) to assess the gene expression profiles and immune infiltration of these tumors. We will also compare the TCGA breast primary with a paired metastatic breast cancer dataset to see if the trend we observe in the primary tumors also holds true in metastatic tumors.

Disease Background:

- Breast Cancer Molecular Subtypes and Metastasis

- Overview of Luminal: Hormone (ER) and/or HER2 positive tumors with better survival than other breast cancer subtypes. Luminal A tumors are the most hormone (ER) and/or HER2 positive and have the best survival among all breast cancer subtypes. Luminal B tumors are more aggressive than Luminal A tumors, but still have a better survival than other breast cancer subtypes.
  - Luminal A: ER+ and/or PR+, HER2-, Ki67 low
  - Luminal B: ER+ and/or PR+, HER2+ or HER2-, Ki67 high
- Most common metastasis sites: bone, lung, liver, brain
- Luminal metastasis: Luminal A tumors are more likely to develop bone metastasis, and Luminal B tumors are more likely to develop lung and liver metastasis. Luminal A tumors are also more likely to develop late metastasis (>5 years after initial diagnosis) than other breast cancer subtypes.
- Immune Response in Cancer
  - Immune response plays a dual role in cancer development and progression. Immune cells can suppress tumor growth and prevent metastasis, but they can also promote tumor growth and metastasis by suppressing anti-tumor immunity and promoting inflammation.
  - Tumor infiltration by immune cells is a strong predictor of survival in many cancers, including breast cancer. Tumors with more immune cell infiltration generally have better survival outcomes.
  - Tumors with higher expression of immune response genes (IRG) also tend to have better survival outcomes in many cancers, including breast cancer.
- Immune Response in Breast Cancer Metastasis
  - Immune response in metastatic breast cancer is complex and can vary depending on the metastatic site. For example, lung metastasis may have a different immune response than bone metastasis.
  - Understanding the immune response in metastatic breast cancer is important for developing effective immunotherapy strategies for these patients.
- Immunotherapy: Pembrolizumab (anti-PD1) approved for PD-L1+ metastatic TNBC
- SWOG ongoing trial: Pembrolizumab + chemotherapy in luminal metastatic breast cancer (S1416)
- Immune Response in Luminal Metastatic Breast Cancer
  - Luminal A tumors have a low immune response and are less likely to respond to immunotherapy.
  - Luminal B tumors have a higher immune response and may be more responsive to immunotherapy.
  - Understanding the immune response in luminal metastatic breast cancer is important for developing effective immunotherapy strategies for these patients.
- Tumor Immune Microenvironment (TIME) in Breast Cancer Metastasis
  - The TIME can vary between primary tumors and metastatic sites, and it can also change over time during treatment and progression.
  - Understanding the TIME in metastatic breast cancer is important for developing effective immunotherapy strategies.
- Relevant Genes
  - PD-L1 (CD274): Ligand for PD-1, a key immune checkpoint protein. Tumors with high PD-L1 expression may be more susceptible to immunotherapy.
  - CD8A: Marker for cytotoxic T cells, which are immune cells that can kill tumor cells.
  - FOXP3: Marker for regulatory T cells (Tregs), which are immune cells that can suppress the immune response.
  - CD68: Marker for macrophages, which can be both pro-tumor and anti-tumor.
  - IFNG: Interferon-gamma, a cytokine that can promote immune response.
  - GZMB: Granzyme B, a protease that can induce apoptosis in tumor cells.
- Relevant Pathways
  - Interferon signaling pathway: Important for immune response to viral infections and cancer.
  - Antigen presentation pathway: Important for the immune response to cancer.
  - Cytokine signaling pathway: Cytokines are signaling molecules that can influence the immune response.
- Relevant Tools
  - TIMER: A web server for the comprehensive analysis of tumor infiltrating immune cells.
  - cBioPortal: A web resource for exploring multi-dimensional cancer genomics data.
  - GEPIA: A web server for cancer and normal gene expression profiling and interactive analyses.
  - UCSC Xena: A platform for exploring and analyzing multi-omic and clinical data.
- Relevant Datasets
  - TCGA BRCA: The Cancer Genome Atlas Breast Invasive Carcinoma dataset.
  - GTEx: The Genotype-Tissue Expression project, a resource for studying gene expression and regulation in various tissues.
  - Metabric: The Molecular Taxonomy of Breast Cancer International Consortium dataset.
  - Kim et al. 2018: A study on the immune landscape of breast cancer metastasis to various organs.
  - Savas et al. 2018: A study on the immune landscape of breast cancer metastasis to the lymph node.
- Hypothesis & Expectations
  - The role of IRG in prognosis may be different in luminal metastatic breast cancer compared to primary breast cancer.
  - Luminal A tumors with high IRG expression may be more likely to have better survival outcomes than those with low IRG expression.
  - Luminal B tumors with high IRG expression may be more likely to have better survival outcomes than those with low IRG expression, due to their higher immune response.
  - The immune response in metastatic breast cancer may be different from that in primary breast cancer, and this difference may affect the prognostic value of IRG.
  - The immune response in luminal metastatic breast cancer may be different from that in other metastatic breast cancer subtypes, and this difference may affect the prognostic value of IRG.

Data Sets:

We will use the Cancer Genome Atlas (TCGA) and Tumor Immune Estimation Resource (TIMER) to assess the gene expression profiles and immune infiltration of these tumors. We will also compare the TCGA breast primary with a paired metastatic breast cancer dataset to see if the trend we observe in the primary tumors also holds true in metastatic tumors.

- TCGA Breast Cancer Primary Tumor Data
  - Clinical data: Patient demographics, tumor characteristics, treatment information, survival outcomes.
  - Gene expression data: RNA-seq data to quantify gene expression levels.
  - Immune cell infiltration data: Estimates of immune cell infiltration from various deconvolution methods.
- TCGA Breast Cancer Metastatic Tumor Data
  - Clinical data: Patient demographics, tumor characteristics, treatment information, survival outcomes.
  - Gene expression data: RNA-seq data to quantify gene expression levels.
  - Immune cell infiltration data: Estimates of immune cell infiltration from various deconvolution methods.
- Paired Primary and Metastatic Tumor Data
  - Gene expression data: RNA-seq data to compare gene expression between primary and metastatic tumors.
  - Immune cell infiltration data: Estimates of immune cell infiltration in primary and metastatic tumors.
- Immune Response Gene Signatures
  - Interferon-gamma (IFN-γ) signature: Genes induced by interferon-gamma, a cytokine that can promote immune response.
  - T cell inflamed signature: Genes expressed in tumors with a T cell inflamed phenotype, indicating an immune response.
  - Cytolytic activity signature: Genes related to the cytolytic activity of immune cells, such as cytotoxic T cells and natural killer cells.
  - Custom signature: We can also create a custom signature by combining genes from different pathways or literature.
- Other Relevant Datasets
  - METABRIC: A large breast cancer dataset with clinical and molecular data.
  - GTEx: Gene expression data from various tissues, including breast tissue.
  - Kim et al. 2018: A study on the immune landscape of breast cancer metastasis to various organs.
  - Savas et al. 2018: A study on the immune landscape of breast cancer metastasis to the lymph node.

Data Analysis:

Methods

The machine learning technique I want to do this project is Principal Component Analysis (PCA). PCA is an unsupervised linear dimensionality reduction technique that can be used to visualize high-dimensional data and identify patterns in the data. We expect the most important variation in the gene expression data to be captured by the first few principal components. We will use PCA to visualize the expression of interferon-related genes (IFNG, STAT1, IRF1) among breast cancer samples. We will examine the PC loadings to see which genes contribute most to the observed variation and assess whether samples with high expression of these genes cluster together. We will also perform PCA separately for primary and metastatic samples to see if the patterns of gene expression and immune infiltration are consistent across different tissue sites. We will use PCA to see if the samples with high immune signatures (IFN-γ signature, T cell inflamed signature, cytolytic activity signature) cluster together and have lower tumor purity, which would indicate higher immune infiltration. We will also use PCA to see if the samples with high immune scores from different deconvolution methods cluster together, which would indicate that the different methods are capturing similar patterns in the immune cell infiltration data.

Results

```
# Load packages and environment
library(tidyverse)
library(ggplot2)
library(dplyr)
library(tibble)
library(stringr)
library(pheatmap)
library(RColorBrewer)
library(ggfortify)
library(ggrepel)
library(tibble)
library(tidyverse)
library(ggplot2)
library(dplyr)
library(stringr)
library(pheatmap)
library(RColorBrewer)
library(ggfortify)
library(ggrepel)
library(tibble)
# Set working directory
setwd("~/Desktop/Data_Science/Module_3/Module_3_Project")
# Create directory structure
dir.create("data", recursive = TRUE)
dir.create("results", recursive = TRUE)
dir.create("results/figures", recursive = TRUE)
dir.create("results/tables", recursive = TRUE)
# Check current directory and structure
getwd()
list.files("data")
list.files("results")
# Load the TCGA BRCA expression data
brca_expr <- read.csv("data/TCGA_BRCA_expression.csv")
# Check the dimensions and first few rows
dim(brca_expr)
head(brca_expr[, 1:5])
# Check for any missing values
sum(is.na(brca_expr))
# Check the sample types
table(substr(colnames(brca_expr)[-1], 13, 15))
# Create a metadata frame with sample IDs and sample types
metadata <- data.frame(
  sample = colnames(brca_expr)[-1],
  type = substr(colnames(brca_expr)[-1], 13, 15)
)
# Check the metadata
head(metadata)
# Create a logical vector for primary tumors
is_primary <- metadata$type == "01A"
# Check how many primary tumors we have
sum(is_primary)
# Create a new expression matrix with only primary tumors
brca_expr_primary <- brca_expr[, c(1, which(is_primary) + 1)]
# Check the dimensions of the new matrix
dim(brca_expr_primary)
# Define interferon-related genes
ifn_genes <- c("IFNG", "STAT1", "IRF1")
# Check if these genes are in our data
ifn_genes %in% brca_expr_primary$X
# Extract expression data for these genes
ifn_expr <- brca_expr_primary[brca_expr_primary$X %in% ifn_genes, ]
# Check the dimensions
dim(ifn_expr)
# Look at the first few rows
head(ifn_expr[, 1:5])
# Transpose the data for PCA
ifn_expr_t <- t(ifn_expr[, -1])
colnames(ifn_expr_t) <- ifn_expr$X
# Check the dimensions
dim(ifn_expr_t)
# Look at the first few rows
head(ifn_expr_t[, 1:5])
# Perform PCA
ifn_pca <- prcomp(ifn_expr_t, scale. = TRUE)
# Check the PCA object
summary(ifn_pca)
# Create a data.frame for plotting
ifn_pca_df <- data.frac(ifn_pca$x)
# Add sample type information
ifn_pca_df$type <- metadata[rownames(ifn_pca_df), "type"]
# Check the new data.frame
head(ifn_pca_df)
# Create the plot
ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
ifn_pca_df$type <- factor(ifn_pca_df$type, levels = c("01A", "11A"))
ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
ifn_pca_df <- data.frame(ifn_pca_df)
ifn_pca_df$type <- factor(ifn_pca_df$type, levels = c("01A", "11A"))
ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
ifn_pca_df <- as.data.frame(ifn_pca_df)
ifn_pca_df$type <- factor(ifn_pca_df$type, levels = c("01A", "11A"))
ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
ifn_pca_df <- ifn_pca$x
ifn_pca_df$type <- metadata[rownames(ifn_pca_df), "type"]
ifn_pca_df$type <- factor(ifn_pca_df$type, levels = c("01A", "11A"))
ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
# Create the plot
ggplot(ifn_pca_df, aes(PC1, PC2, color = type))
Error: Cannot find `PC1`, `PC2`, `type` or `.pairs` in `.data`.
# Check the column names
colnames(ifn_pca_df)
# Create the plot with the correct column names
ggplot(ifn_pca_df, aes(PC1, PC2, color = type))
Error: Cannot find `PC1`, `PC2`, `type` or `.pairs` in `.data`.
# Create a simple plot
plot(ifn_pca_df$PC1, ifn_pca_df$PC2, col = ifn_pca_df$type)
# Add a legend
legend("topright", legend = levels(ifn_pca_df$type), col = 1:2, pch = 1)
# Create a more publication-ready plot
pdf("results/figures/ifn_pca.pdf")
plot(ifn_pca_df$PC1, ifn_pca_df$PC2, col = ifn_pca_df$type,
     main = "PCA of BRCA samples (Interferon genes)",
     xlab = "PC1 (52.7% variance)",
     ylab = "PC2 (17.1% variance)")
legend("topright", legend = levels(ifn_pca_df$type), col = 1:2, pch = 1)
dev.off()
# Create a boxplot for PC1 values by sample type
boxplot(ifn_pca_df$PC1 ~ ifn_pca_df$type,
        main = "PC1 by sample type",
        xlab = "Sample type",
        ylab = "PC1")
# Create a boxplot for PC2 values by sample type
boxplot(ifn_pca_df$PC2 ~ ifn_pca_df$type,
        main = "PC2 by sample type",
        xlab = "Sample type",
        ylab = "PC2")
# Create a directory for the results
dir.create("results", recursive = TRUE)
# Save the PCA plot as a PDF
pdf("results/ifn_pca.pdf")
plot(ifn_pca_df$PC1, ifn_pca_df$PC2, col = ifn_pca_df$type,
     main = "PCA of BRCA samples (Interferon genes)",
     xlab = "PC1 (52.7% variance)",
     ylab = "PC2 (17.1% variance)")
legend("topright", legend = levels(ifn_pca_df$type), col = 1:2, pch = 1)
dev.off()
# Save the boxplots
pdf("results/ifn_pca_boxplots.pdf")
boxplot(ifn_pca_df$PC1 ~ ifn_pca_df$type,
        main = "PC1 by sample type",
        xlab = "Sample type",
        ylab = "PC1")
boxplot(ifn_pca_df$PC2 ~ ifn_pca_df$type,
        main = "PC2 by sample type",
        xlab = "Sample type",
        ylab = "PC2")
dev.off()
# Check the directory structure
list.files("results")
list.files("results/figures")
# Load the metadata
metadata <- read.csv("data/TCGA_BRCA_metadata.csv")
# Check the dimensions and first few rows
dim(metadata)
head(metadata[, 1:5])
# Check the column names
colnames(metadata)
# Create a new column for the sample type
metadata$type <- substr(metadata$X, 13, 15)
# Check the new column
table(metadata$type)
# Create a logical vector for primary tumors
is_primary <- metadata$type == "01A"
# Check how many primary tumors we have
sum(is_primary)
# Create a new expression matrix with only primary tumors
brca_expr_primary <- brca_expr[, c(1, which(is_primary) + 1)]
# Check the dimensions of the new matrix
dim(brca_expr_primary)
# Define interferon-related genes
ifn_genes <- c("IFNG", "STAT1", "IRF1")
# Check if these genes are in our data
ifn_genes %in% brca_expr_primary$X
# Extract expression data for these genes
ifn_expr <- brca_expr_primary[brca_expr_primary$X %in% ifn_genes, ]
# Check the dimensions
dim(ifn_expr)
# Look at the first few rows
head(ifn_expr[, 1:5])
# Transpose the data for PCA
ifn_expr_t <- t(ifn_expr[, -1])
colnames(ifn_expr_t) <- ifn_expr$X
# Check the dimensions
dim(ifn_expr_t)
# Look at the first few rows
head(ifn_expr_t[, 1:5])
# Perform PCA
ifn_pca <- prcomp(ifn_expr_t, scale. = TRUE)
# Check the PCA object
summary(ifn_pca)
# Create a data.frame for plotting
ifn_pca_df <- data.frac(ifn_pca$x)
# Add sample type information
ifn_pca_df$type <- metadata[rownames(ifn_pca_df), "type"]
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ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
ifn_pca_df <- as.data.frame(ifn_pca_df)
ifn_pca_df$type <- factor(ifn_pca_df$type, levels = c("01A", "11A"))
ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
ifn_pca_df <- ifn_pca$x
ifn_pca_df$type <- metadata[rownames(ifn_pca_df), "type"]
ifn_pca_df$type <- factor(ifn_pca_df$type, levels = c("01A", "11A"))
ggplot(ifn_pca_df, aligned = TRUE, ...)
Error in UseMethod("aligned") : 
  no applicable method for 'aligned' applied to an object of class "c('matrix', 'numeric')"
# Create the plot
ggplot(ifn_pca_df, aes(PC1, PC2, color = type))
Error: Cannot find `PC1`, `PC2`, `type` or `.pairs` in `.data`.
# Check the column names
colnames(ifn_pca_df)
# Create the plot with the correct column names
ggplot(ifn_pca_df, aes(PC1, PC2, color = type))
Error: Cannot find `PC1`, `PC2`, `type` or `.pairs` in `.data`.
# Create a simple plot
plot(ifn_pca_df$PC1, ifn_pca_df$PC2, col = ifn_pca_df$type)
# Add a legend
legend("topright", legend = levels(ifn_pca_df$type), col = 1:2, pch = 1)
# Create a more publication-ready plot
pdf("results/figures/ifn_pca.pdf")
plot(ifn_pca_df$PC1, ifn_pca_df$PC2, col = ifn_pca_df$type,
     main = "PCA of BRCA samples (Interferon genes)",
     xlab = "PC1 (52.7% variance)",
     ylab = "PC2 (17.1% variance)")
legend("topright", legend = levels(ifn_pca_df$type), col = 1:2, pch = 1)
dev.off()
# Create a boxplot for PC1 values by sample type
boxplot(ifn_pca_df$PC1 ~ ifn_pca_df$type,
        main = "PC1 by sample type",
        xlab = "Sample type",
        ylab = "PC1")
# Create a boxplot for PC2 values by sample type
boxplot(ifn_pca_df$PC2 ~ ifn_pca_df$type,
        main = "PC2 by sample type",
        xlab = "Sample type",
        ylab = "PC2")
# Save the boxplots
pdf("results/figures
```

```
            f'Chi²={chi2:.2f}, p={p_val:.4f}',
            ha='center', fontsize=10)
    plt.tight_layout()
    plt.show()

    # ------------------------------
    # 1️⃣3️⃣ Age Grouping + Chi-square Test
    # ------------------------------
    bins = [0, 40, 60, 80, 120]
    labels = ['<40', '40-60', '60-80', '>80']
    expr_telomere_meta['Age_Group'] = pd.cut(expr_telomere_meta['age_at_diagnosis'], bins=bins, labels

    age_crosstab = pd.crosstab(expr_telomere_meta['Age_Group'], expr_telomere_meta['Telomerase_Group']
    print("\nCrosstab of Age Group vs Telomerase Group:")
    print(age_crosstab)

    chi2_age, p_age, dof_age, exp_age = chi2_contingency(age_crosstab)
    print(f"Chi-square (Age Groups vs Telomerase Group): chi2={chi2_age:.3f}, p-value={p_age:.4f}")

    # Bar plot for categorical age distribution
    age_crosstab.plot(kind='bar', figsize=(6,5), colormap='Set2')
    plt.title('Telomerase Group Distribution Across Age Groups')
    plt.xlabel('Age Group')
    plt.ylabel('Number of Samples')
    plt.legend(title='Telomerase Group')
    plt.tight_layout()
    plt.show()

else:
    print("⚠️ 'age_at_diagnosis' not found in metadata.")
```

```
BRCA metadata shape: (80, 72)
BRCA expression shape: (15716, 80)
Total genes in BRCA dataset: 15716
Telomerase-related genes present: ['TERT', 'POT1', 'MYC', 'DKC1']
Number of telomerase genes present: 4
Training samples: 56, Test samples: 24
PC1 separation t-test: t=8.900, p=0.0000
Silhouette score (group separation): 0.320
```

```
Top genes contributing to PC1:
MYC     0.610029
DKC1    0.571239
TERT    0.536889
POT1    0.115329
Name: PC1_loading, dtype: float64

Top genes contributing to PC2:
POT1    0.901596
DKC1    0.195638
MYC    -0.014328
TERT   -0.385545
Name: PC2_loading, dtype: float64

T-test (Age vs Telomerase Group): t=-0.255, p-value=0.7997
/var/folders/dq/kzp57cm57n30vpnnzf3ktvgm0000gn/T/ipykernel_58844/1164561190.py:160: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x`
variable to `hue` and set `legend=False` for the same effect.

  sns.violinplot(
```

```
Crosstab of Age Group vs Telomerase Group:
Telomerase_Group  High  Low
Age_Group
<40                  1    2
40-60               21   17
60-80               15   16
>80                  2    2
Chi-square (Age Groups vs Telomerase Group): chi2=0.735, p-value=0.8651
```

Verify and validate your analysis:

After reducing the BRCA telomerase gene expression data to 2 PCs, we measure how well the original gene expression is reconstructed from the PCs. Our PCA plot of telomerase genes captures about ~70% of variance, meaning that studying telomerase genes is a good factor to study at as it can account for a lot variance and can possibly explain some possible clincal values using telomerase gene expression levels. Comparing PCA results between training and test datasets revealed consistent data trends. While the test set exhibited slightly higher variance along the principal components, the distributions of PC scores were statistically indistinguishable, with KS test having a p value greater than 0.23. Principal components showed strong alignment, with a cosine similarity score of greater than 0.8, confirming stable and reproducible dimensionality reduction across datasets.

From our results from above we can see a clear divide between high expression of Telomerase genes and low expression of Telomerase genes from our PCA model. From that model we can also tell that MYC and TERT are some of the most important telomerase genes contributing to our PCA model as they show to have the highest contribution while POT1 has less. From our violin plot regarding expression of telomerase genes and age of diagnosis, it shows that there is so significant correlation between level of expression of telomerase genes and age of diagnosis. Our chi-square test with a result of chi2 = 0.735 shows that that the expression of telomerase genes and age of diagnosis are independent of each other.

External Validation:

Our results are supported by findings from a large pan-cancer study by Luo and others, who showed that tumors with higher telomerase activity often behave differently at the molecular level. In their analysis, telomerase-high tumors tended to group together, showing stronger proliferation signals and distinct gene-expression patterns compared with telomerase-low tumors. This mirrors what we see in our PCA: BRCA tumors with higher expression of telomerase-related genes form their own cluster and are largely driven by genes like MYC and TERT. Seeing the same pattern in a much larger dataset helps reinforce that the separation we observe in our BRCA samples is real and biologically meaningful.

citation:

- Luo, Z., Wang, W., Li, F., Songyang, Z., Feng, X., Xin, C., Dai, Z., & Xiong, Y. (2019). Pan-cancer analysis identifies telomerase-associated signatures and cancer subtypes. Molecular Cancer, 18(1), 106. https://doi.org/10.1186/s12943-019-1035-x

Conclusions and Ethical Implications:

Conclusion:

- Overall, our results suggest that differences in telomerase-related gene expression do matter in BRCA tumors. The PCA shows that telomerase-high and telomerase-low samples form noticeably different patterns, which helps answer our main question. PC1 is driven mostly by genes like TERT, MYC, and DKC1, highlighting how telomerase activity and proliferation shape the main variation in this dataset. While there is still some overlap between groups, the separation we do observe indicates that telomerase expression levels contribute meaningfully to tumor behavior. The strong alignment between the training and test data further supports that these patterns are real and reliable, rather than random variation.

Ethical implications:

- Gene-expression results need to be interpreted with care—telomerase activity on its own isn't enough to predict how any one patient will do without considering their full clinical picture.
- Our findings should ultimately be used to improve care for all groups, not widen gaps, especially because access to genomic testing and precision treatments is not equal across communities.

Limitations and Future Work:

Future Work:

- Explore how telomerase-related expression changes across different stages of tumor progression Test additional methods to capture non-linear patterns in telomere biology.
- Examine whether telomerase activity patterns differ between younger and older patients, since age can influence telomere biology.

limitations:

- Limitations Our analysis focuses on a small set of telomerase-related genes, which may miss other biological factors influencing tumor behavior.
- PCA captures only linear patterns, so more complex or non-linear relationships in the data may not be fully represented.
- The telomerase-high and telomerase-low groups were defined using expression cutoffs, which may simplify the true biological variation.

NOTES FROM YOUR TEAM:

This is where our team is taking notes and recording activity.

QUESTIONS FOR YOUR TA:

These are questions we have for our TA.